



Technical Service Bulletin

Technical Service Bulletin: TSB230092

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Fan Clutch Oil Supply Manifold Solenoid Re-orientation

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Warranty Statement

The information in this document has no effect on present warranty coverage or repair practices, nor does it authorize TRP or Campaign actions.

Contents

Product Affected

- QSK50 CM2150 MCRS
- QSK50 CM850 MCRS
- QSK50 CM2350 K108
- QSK50 CM2350 K130

Issue

Fan clutch oil solenoid coil malfunction causes permanent cooling fan engagement. Fault codes 245 and 2377 become active. Permanent full fan speed can cause over-cooling and excessive noise.

Verification

- Engine cooling fan runs constantly
- Fault codes 245 and 2377 become active

Resolution

A new fan clutch oil manifold has been designed to improve oil cooling on the oil control solenoid coil. This fan clutch oil manifold is released in a parts kit (part number 6468466). The parts kit contains all the required components to upfit the new manifold and separately available solenoid (part number 3641305) to the engine.

Existing design oil manifold:

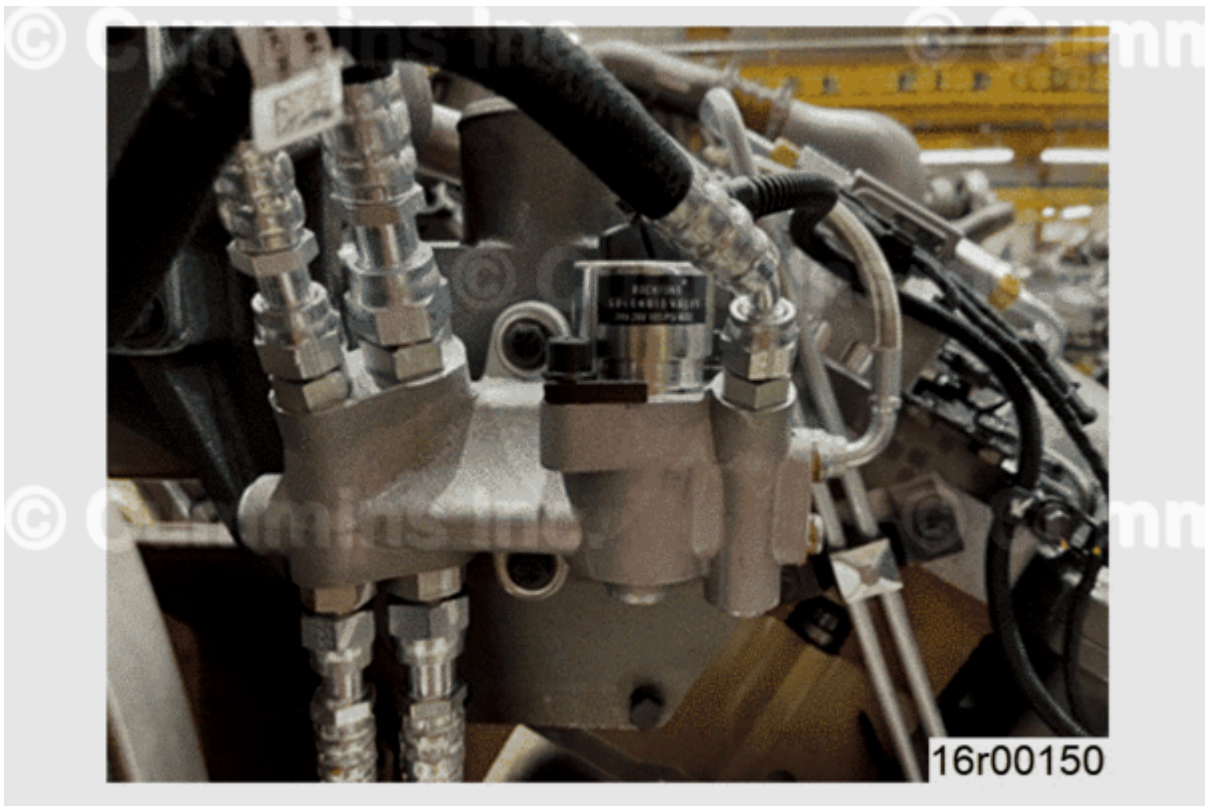


Figure 1, Existing Oil Manifold Installation.

New design oil manifold:

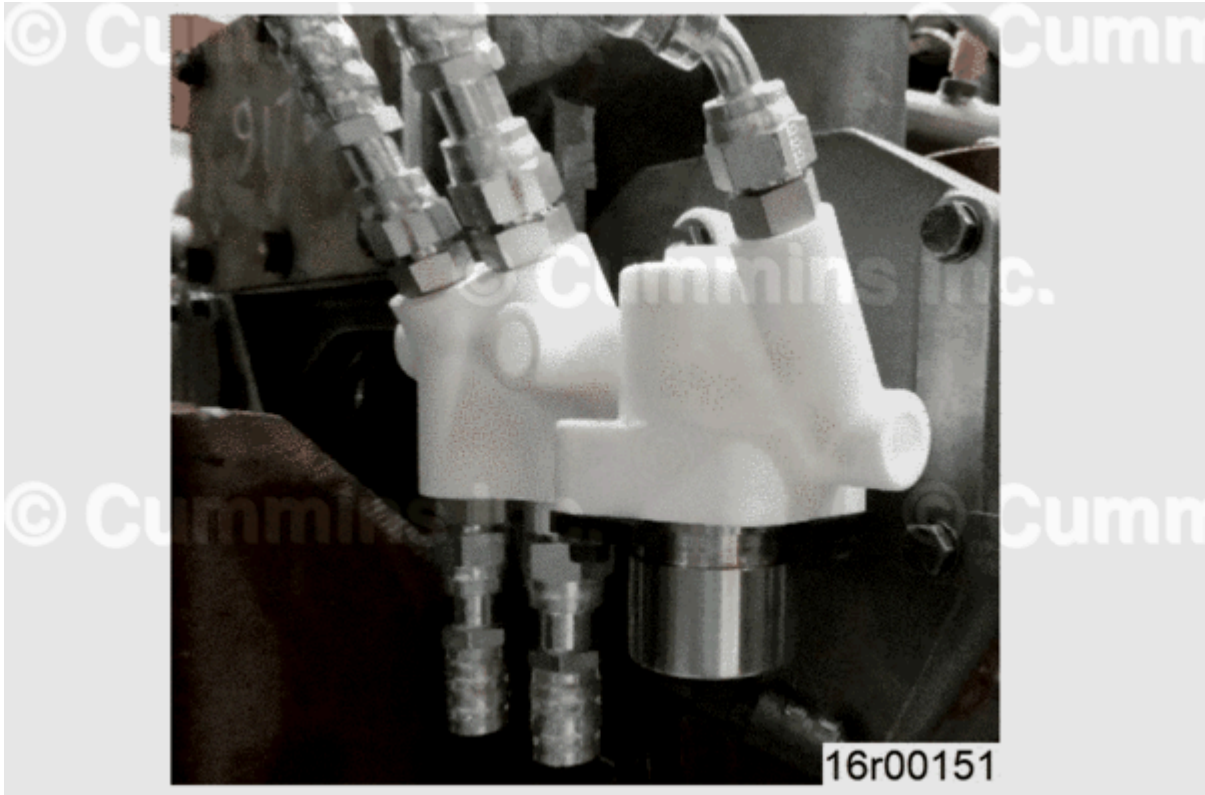


Figure 2, New Design Oil Manifold Installation.

Reason for Change

The new design oil manifold rotates the solenoid mounting location by 180°. In this orientation, cooling oil is gravity fed to the solenoid coil, reducing coil temperature and therefore the risk of malfunction.

Service Instructions

Order upfit kit, Part Number 6468466 and solenoid valve, Part Number 3641305, before attempting to carry out this repair. Check part quantities delivered in the parts kit, as per Table 1, Service Parts. Remove the five male – male unions/ adapters from the existing manifold. Re-use these unions/ connectors. O-rings will be provided in the parts kit.

To replace the fan clutch oil control manifold and solenoid, see corresponding Service Manual. Reference Procedure 007-125 in Section 7.

- GTA38, K38, K50 QSK38 and QSK50 Service Manual, Bulletin 4021528
- QSK50 CM2350 K108 Service Manual, Bulletin 4332823
- QSK50 CM2350 K130 Service Manual, Bulletin 5504180

Following installation of the fan clutch oil control manifold and solenoid, the solenoid wiring harness **must** be re-routed to allow connector alignment with the new solenoid position. This process differs for Tier 2 and Tier 4 engines. Verify the route required and re-route as shown in Figure 5.

QSK50 CM2150 MCRS Engines:

- Remove the solenoid harness from the shared P-clip, holding the solenoid harness and the engine harness, to the rocker cover (circled yellow). Cut the cable tie securing the solenoid harness to the existing engine harness. Figure 3 shows the existing route for Tier 2 engines.

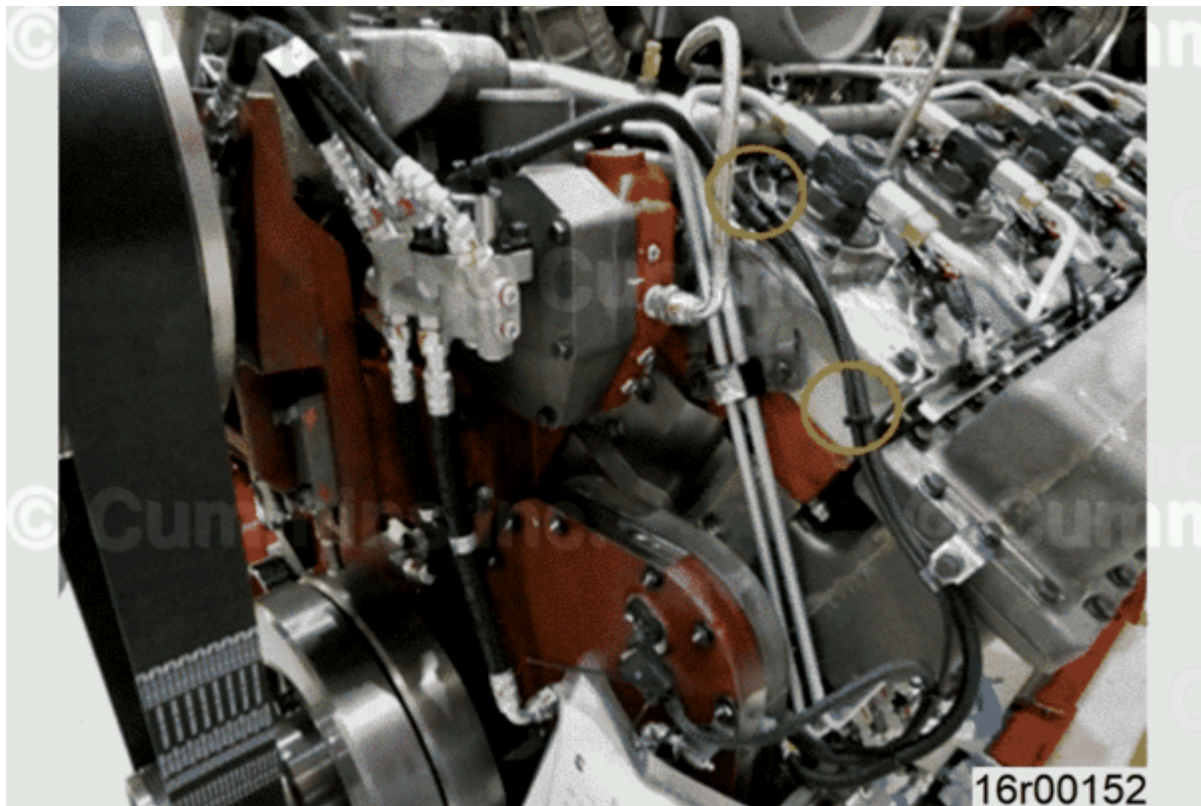


Figure 3, Tier 2 Engine Wiring Harness Routing.

QSK50 CM2350 K108/ K130 Engines:

- Figure 4 shows the existing Tier 4 solenoid harness route, which has an engine harness P-clipped to it on the rocker cover. Remove the solenoid harness from the engine harness P-clip.

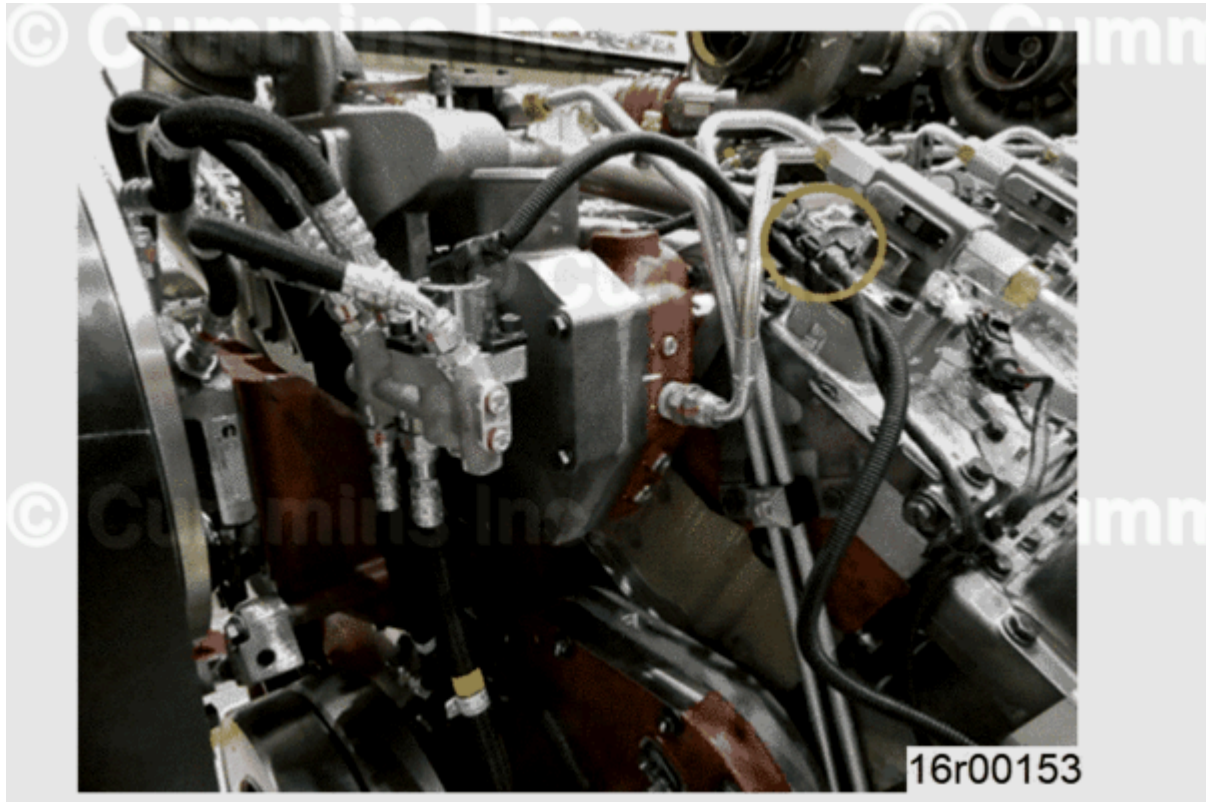


Figure 4, Tier 4 Engine Wiring Harness Route.

All Engines:

- Secure existing Tier 2 and Tier 4 non-solenoid harnesses as per the original installation.
- Re-route the solenoid harness to the new solenoid position. Figure 6 shows the required new route using two P-clips (part number 68152A) on existing gear cover capscrew locations across both Tier 2 and Tier 4 engines.
- Remove two gear cover capscrews and install the solenoid harness through the P-clips, vertically upwards as shown.
- Install the capscrews.

Torque Value: 75 n·m [55 ft-lb]

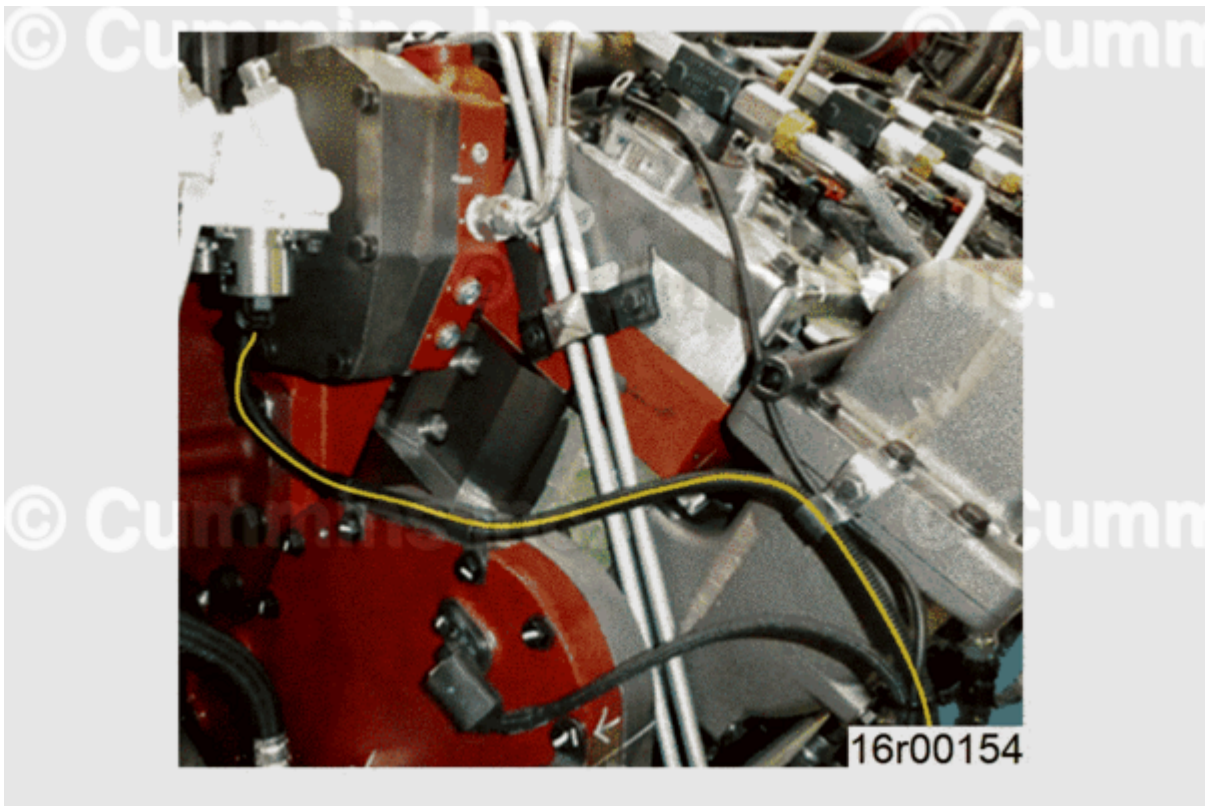


Figure 5, New Solenoid Harness Routing.

- Connect the wiring harness to the solenoid plug. Verify a positive “click” as the connector fork slots over the plug and locks in position to avoid backing out.

Service Parts Availability

Service parts are available in kit form. See Table 1 for part numbers contained within the parts kit.

Table 1, Service Parts					
Part Description	Existing Part Number	Obsolete	Superseded	New Part Number	Part Quantity
MANIFOLD, OIL	3640889	Yes	Yes	6414133	1
SCREW, SOCKET HEAD CAP	S143B	No	No	-	2
WASHER, LOCK	S604	No	No	-	2
GASKET, CORROSION RESISTOR	3650106	No	No	-	1
PLUG, PIPE	3008466	No	No	-	2
PLUG, THREADED	3046202	No	No	-	2
PLUG, THREADED	3037536	No	No	-	1

Table 1, Service Parts					
Part Description	Existing Part Number	Obsolete	Superseded	New Part Number	Part Quantity
CLIP	68152A	No	No	-	2
SEAL, O RING	3348871	No	No	-	2
SEAL, O RING	3627695	No	No	-	2
SEAL, O RING	3089240	No	No	-	2
SEAL, O RING	3630740	No	No	-	2
SEAL, O RING	3922794	No	No	-	1
SEAL, O RING	3037537	No	No	-	1

Part Compatibility

The new oil manifold is common across the QSK50 Tier 2 and Tier 4, as was the existing design. New and existing oil manifolds are backwards compatible. The new manifold is advised for expected fan solenoid operation and service life.

Part Identification

The new oil manifold is different in shape to the existing part, allowing the oil control solenoid to sit upside down. The new part number is stamped in the location shown in Figure 6 The existing part number is stamped exactly 180° opposite to this on the bottom face. Both part numbers are stamped on the face opposite to the solenoid when viewed as installed.

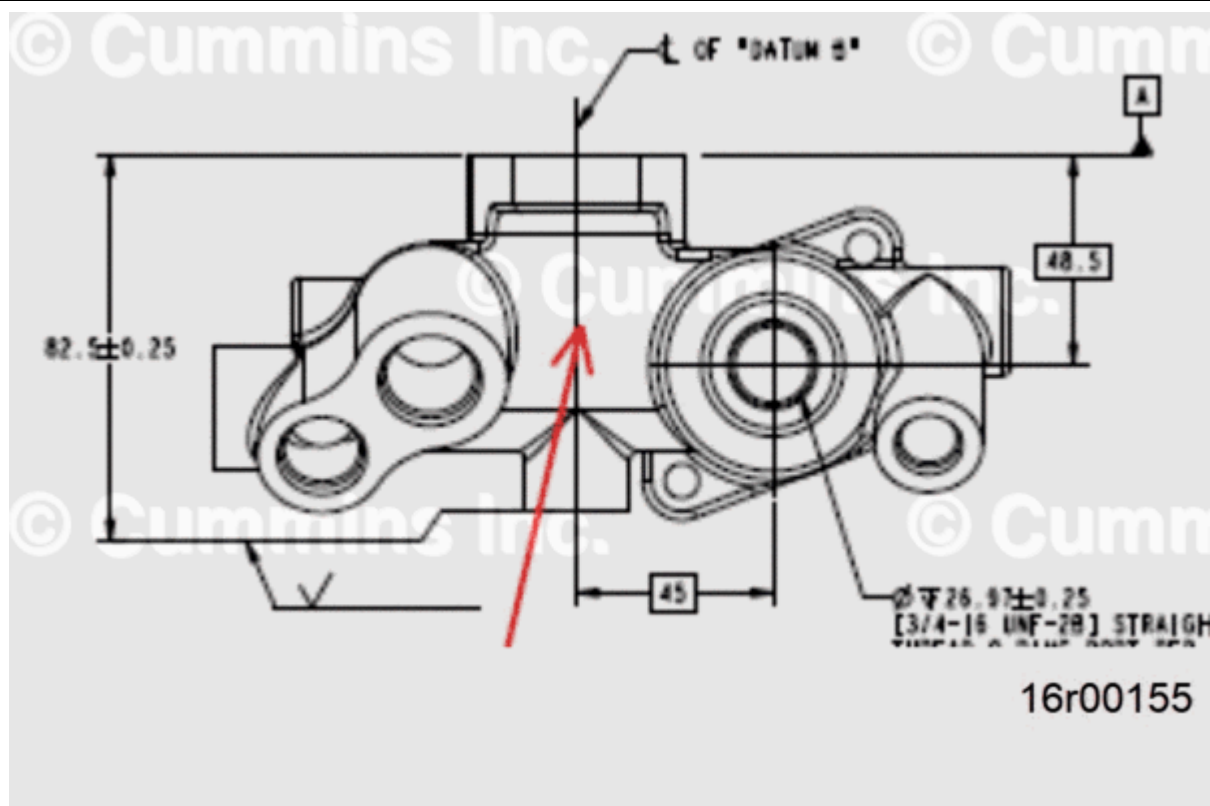


Figure 6, Part Number Stamping Location on New Fan Clutch Oil Manifold.

Part Inventory Action

Obsolete part numbers in stock are **not** to be used.

Production Status

Implemented for production. See Table 2.

Table 2, Production Information

ESN First	Build Date ¹	Plant
33231741	8 April 2023	Daventry Engine Plant

¹Engine build date can be found on engine dataplate.

Publications Affected

Table 3, Publications Affected

Manual Type	Engine	Bulletin Number	Procedure Title	Procedure	Section
Service Manual	QSK50 CM2150 MCRS	4021528	Fan Clutch Lubricating Oil Manifold	Refer to Procedure 007-125 (/qs3/pubsys2/xml/en/procedures/28/28-007-125.html)	7

Table 3, Publications Affected					
Manual Type	Engine	Bulletin Number	Procedure Title	Procedure	Section
Service Manual	QSK50 CM2350 K130	5504180	Fan Clutch Lubricating Oil Manifold	Refer to Procedure 007-125 (/qs3/pubsy s2/xml/en/pr ocedures/53 6/536-007- 125.html)	7
Service Manual	QSK50 CM2350 K108	4332823	Fan Clutch Lubricating Oil Manifold	Refer to Procedure 007-125 (/qs3/pubsy s2/xml/en/pr ocedures/25 5/255-007- 125.html)	7

Document History

Date	Details
2023-4-25	Module Created